

B.TECH (ECE)
III YEAR, VI SEMESTER EXAMINATION
DIGITAL SIGNAL PROCESSING

[Time : 03:00 Hours]

[Max. Marks : 70]

NOTE:- Attempt all the questions as per given instruction. Use of Scientific Calculator is allowed.

Ques:-1 Attempt any Four parts of this question:

(5*4 = 20)

- Determine the z-Transform of $x(n) = 2^n u(n-2)$
- Determine the convolution of the signal given below using ZT
 $x_1(n) = (1/2)^n u(n)$, $x_2(n) = 3^n u(-n)$.
- Write a short note on Sampling Theorem.
- Define the Recursive and Non-Recursive structures.
- Explain the Causal and Non-causal Systems with examples.
- Find the DFT of $x(n) = \{ 1/5, 1/5, 1/5 \}$.

Ques:-2 Attempt any Two parts of this question :

(7*2=14)

- Design a high pass filter using Hamming window with a cut-off frequency of 1.2 rad/sec and $N = 9$.
- Obtain the system functions of normalized Butterworth filters for order $N = 1$ & 3.
- Convert the analog filter into digital filter whose system response is

$$H(s) = \frac{1}{(s+0.2)^2 + 16}$$

Using Bilinear Transformation.

Ques:-3 Attempt any Two parts of question.

(8*2=16)

- Design a high pass filter with a frequency response

$$H(e^{j\omega}) = \begin{cases} 1 & \text{for } \pi/4 \leq |\omega| \leq \pi \\ 0 & \text{for } |\omega| \leq \pi/4 \end{cases}$$

Find the value of $h(n)$ for $N=11$ using Hanning Window Technique.

- Find the order of the filter and $H(s)$ of the filter if a digital Chebyshev Filter to satisfy the constraints

$$0.707 \leq |H(e^{j\omega})| \leq 1, \quad \text{for } 0 \leq \omega \leq 0.2\pi$$

$$|H(e^{j\omega})| \leq 0.1, \quad \text{for } 0.5\pi \leq \omega \leq \pi$$

assuming $T=1s$.

- What are different window functions. Write mathematical expression for any Four Window functions.

Ques:-4 Attempt any one part of this question :

(10*1=10)

- Given $x[n] = \{1, 2, 3, 4, 4, 3, 2, 1\}$, find $X[k]$ using DIT FFT algorithm.
- Find the IDFT of $X(k) = \{1, 2, 3, 4\}$

Ques:-5 Attempt any one part of this question:

(10*1=10)

- Given $x[n] = \{1, 2, 3, 4, 5, 6, 7, 8\}$, find $X[k]$ using DIF FFT algorithm.
- Design a Digital Butterworth Filter that satisfying the following constraints

$$0.9 \leq |H(e^{j\omega})| \leq 1; \text{ for } 0 \leq \omega \leq \frac{\pi}{2}$$

$$|H(e^{j\omega})| \leq 0.2; \text{ for } \frac{3\pi}{4} \leq \omega \leq \pi$$

With $T=1s$. Apply Bilinear transformation.
